

ASG Airlines: End to End Data Engineering & Business Analytics Case Study

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Course: MCA

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1. Project Overview

In this Project build an end to end Data Engineering pipeline and Power BI dashboard for ASG airlines a national wide carrier who operates flights in major cities in India. Ingested operation airline data which was provide to us containing flights, payments ,passengers, bookings data. Identifies and handled Data Quality issues such as : missing values ,duplicate records , corrupted identifiers , overnight flights etc. Performed Cleaning ,Standardization, validation check , and transformed the data into a reliable analytical dataset . Also protected the sensitive data using PII masking and equivalent protection method. Prepared the cleaned dataset for business intelligence and reporting . Calculation of operation KPI such as Average Flight Duration , Route- wise Traffic , Delays& Anamolies, Flight Distribution by Airline etc. Build a 4 page Power BI Dashboard to interpret flight operations and supporting data driven decisions .

2. Problem Statement

An Airline runs hundreds of flights daily and there operational data comes from multiple sources such as booking platforms, scheduling systems ,airport logs , payment gateways. However, when the data is pulled into raw excel files the data contains multiple quality issues such as corrupted or inconsistent flights identities, missing values , duplicate values , overnight flights where the arrival occurs the following day. These issues can affect the accuracy of flight duration calculations and make it difficult to generate reliable operational insights . So its important to standardize the data pipeline and other business KPI becomes challenging. Therefore we focus on building a pipeline that can ingest raw data, clean , validate, transform ,standardize the raw data into a clean dataset .

3. Dataset Overview

ASG Airlines contains 4 sheets containing 4,059 raw records

Name	Raw rows	Clean rows	Primary key	Key quality issues	Action taken
Flights	1020	1005	flight_id	15 duplicate rows, 41 missing/unknown airlines, 122 overnight flights	Dropped duplicates, standardized airlines using flight prefix, applied +24h duration correction
Passengers	1039	1000	passenger_id	10 missing last names, PII present	Masked PII and used reporting-safe data
Bookings	1000	1000	booking_id	75 unknown/invalid status values	Standardized to UNKNOWN
Payments	1000	1000	payment_id	78 missing/invalid payment amounts	Converted amounts to numeric and flagged invalid/missing records

4. Solution Architecture

Designed a multi-layer pipeline to transform data into clean business ready reports.

1. Ingestion Layer :
 - Reads 4 different separated sheets (flights, booking, passengers, payments) data from the raw excel sheet provided and ingested it using python pandas.
 - We inspected columns, rows, data types duplicates , missing values , corrupted data.
2. Cleaning & Transformation Layer:
 - Removed only exact duplicate flight records from the flights sheets.
 - Fixed the airline missing column by extracting prefixes from the flight numbers. Eg AI : Air India , 6F : Indigo)
 - Resolved the overnight flight problem by adding 24 hours or 1440 min to negative duration when the arrival data in the following/ next day .
 - Standardize the corrupted booking status into UNKNOWN category .
 - Flags the invalid payment entries without deleting them so the financial audit doesn't mess up .
3. Privacy & Security Layer:
 - Masking sensitive personal information of passengers in file(passengers, booking) like Aadhaar , passport ,emergency contact ,email etc.
 - Created a separate table for (passengers_powerbi) that completely excludes all the PII and ensures zero sensitive data enter the dashboard.
4. Clean Analytical Storage Layer:
 - Exported separate sheets for clean data and removed sensitive data for Power-BI dashboard .
5. BI Reporting Layer:
 - Loaded the clean csv file into power BI desktop.
 - Connects them into Star Schema centered on booking with zero broken links
 - Delivered a 4 page interactive dashboard with filtering and KPI cards.

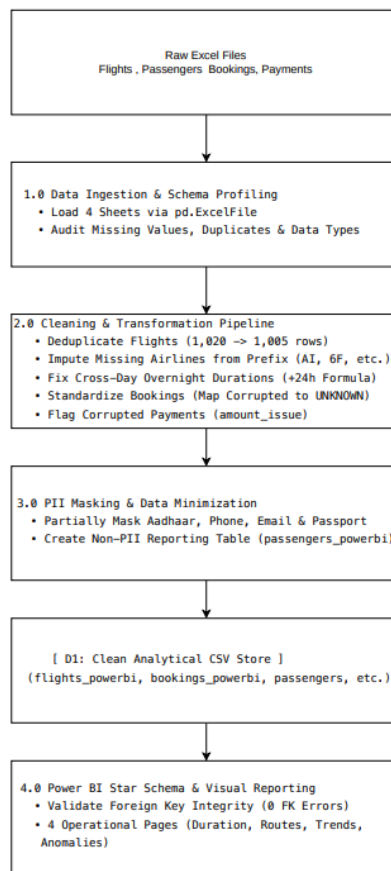


Fig. Architecture Diagram

5. Data Flow Diagram (DFD)

The DFD illustrates the movement of airline data from the raw source files through ingestion, cleaning, validation and PII protection. The resulting reporting-ready data is then used in Power BI for analysis and reporting

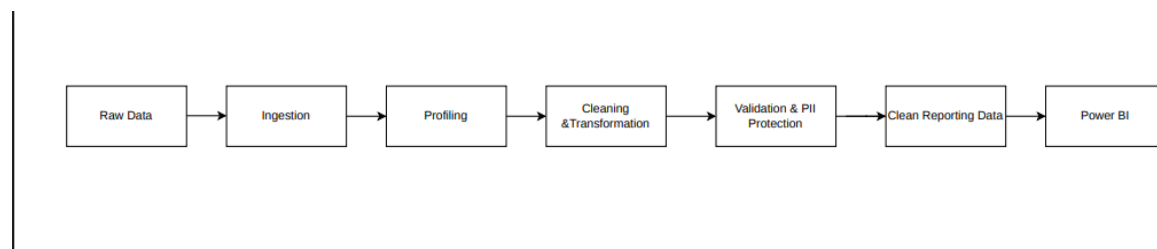


Fig. DFD diagram

6. Data Cleaning & Transformation

Python functions to handle each specific data anomaly found during profiling:

6.1 Flights Dataset Cleaning

1. Exact Duplicate Removal:

- Found 15 exact duplicate rows in the raw flights sheet.
- Dropped duplicates using `flights.drop_duplicates()`, reducing total flights from **1,020 to 1,005 clean records**.

2. Missing Airline Imputation:

- Several flights had missing or unknown airline names, but had flight prefixes like AI, 6E, SG, UK.
- Mapped prefixes to standard carriers (AI → Air India, 6E → IndiGo, SG → SpiceJet, UK → Vistara).

```
# 4. Fill missing / UNKNOWN airline values
# Mapping is based on the patterns present in this dataset
airline_map = {
    "AI": "Air India",
    "6E": "IndiGo",
    "SG": "SpiceJet",
    "UK": "Vistara"
}
```

3. Overnight Flight Duration Fix :

- Identified 122 flights where arrival date was greater than departure date

```
# 5. Convert departure and arrival to datetime
flights_clean["departure_time"] = pd.to_datetime(
    flights_clean["departure_time"],
    errors="coerce"
)

flights_clean["arrival_time"] = pd.to_datetime(
    flights_clean["arrival_time"],
    errors="coerce"
)

# 6. Calculate flight duration in hours
flights_clean["duration_hours"] = (
    flights_clean["arrival_time"] -
    flights_clean["departure_time"]
).dt.total_seconds() / 3600

# 7. Handle overnight flights
# Overnight means the arrival date is after the departure date
flights_clean["overnight_flag"] = (
    flights_clean["arrival_time"].dt.date >
    flights_clean["departure_time"].dt.date
)

# Fix any negative duration caused by incorrect raw dates
flights_clean.loc[
    flights_clean["duration_hours"] < 0,
    "duration_hours"
] += 24

flights_clean["duration_hours"] = (
    flights_clean["duration_hours"].round(2)
)

# 8. Create route
flights_clean["route"] = (
    flights_clean["source"] + " - " +
    flights_clean["destination"]
)

# 9. View the cleaned data
print("First 5 rows after cleaning:")
display(flights_clean.head())
```

- #### 4. Route Column:
- Created a combined route column (source - destination, e.g., DEL - BOM) for easy route-level aggregation in Power BI.

6.2 Passengers Dataset Cleaning

1. **PII Masking:** To protect customer privacy, masked sensitive fields in the operational database.

	passenger_id	email	phone	aadhaar_id
0	P1000	v***	*****3790	*****6001
1	P1001	k***	*****2297	*****2654
2	P1002	m***	*****5092	*****8161
3	P1003	m***	*****7151	*****6475
4	P1004	s***	*****5113	*****6483

2. **Reporting Dataset (passengers_powerbi.csv):** Excluded all PII fields entirely. Kept only passenger_id and non-sensitive demographic details for reporting.

```
# =====  
# PASSENGER REPORTING DATA  
# =====  
  
passengers_reporting = passengers_clean[  
    [  
        "passenger_id",  
        "age",  
        "gender"  
    ]  
].copy()  
  
display(passengers_reporting.head())
```

	passenger_id	age	gender
0	P1000	52	F
1	P1001	15	M
2	P1002	72	M
3	P1003	61	F
4	P1004	21	M

6.3 Bookings Dataset Cleaning

- Corrupted Statuses: 75 records contained corrupted text strings like #ERROR!. Rather than deleting these transactions (which would distort total booking and revenue counts), standardized them to 'UNKNOWN'.
- Final status categories: CONFIRMED, CANCELLED, PENDING, UNKNOWN.

6.4 Payments Dataset Cleaning

- Standardized payment IDs, booking IDs and payment methods.
- Converted payment amounts to numeric values.
- Flagged missing/invalid amounts.
- 922 valid positive payment records and 78 missing/invalid records..

7. Data Quality & Validation

7.1 Validation Summary

Dataset	Validation Check	Result
Flights	Duplicate records	15 exact duplicate rows identified and removed
Flights	Flight duration	Duration calculations validated; overnight cases handled
Passengers	Missing values	10 missing last names identified
Bookings	Status validation	75 records classified as UNKNOWN
Payments	Amount validation	922 valid positive amounts; 78 missing/invalid
Bookings → Flights	Referential integrity	No unmatched flight IDs
Bookings → Passengers	Referential integrity	No unmatched passenger IDs
Payments → Bookings	Referential integrity	No unmatched booking IDs

7.2 Validation Approach

- Duplicate, missing, invalid and inconsistent values were checked after transformation.
- Flight duration calculations and relationships between related datasets were validated.
- Records requiring attention were retained as flagged/UNKNOWN rather than being assigned unsupported values

8. PII Protection & Privacy

8.1 Why Data Privacy Was Necessary

The raw passenger file contained sensitive personal information that can directly identify a person (Personally Identifiable Information - PII). In a real-world company, exposing unmasked Aadhaar cards, passport numbers, personal phone numbers, and emails to analysts or in business dashboards violates data privacy regulations.

Masking the sensitive data:

1. Aadhaar Number: Masked the first 8 digits and kept only the last 4 visible for verification purposes
2. Phone Number: Masked the first 6 digits, showing only the last 4 digits
3. Email Address: Masked the username while preserving the first letter and domain
4. Passport Number: Masked the initial characters, keeping the last 4 visible
5. Emergency Contact: Also masked to prevent exposing family contact numbers.

	passenger_id	email	phone	aadhaar_id
0	P1000	v***	*****3790	*****6001
1	P1001	k***	*****2297	*****2654
2	P1002	m***	*****5092	*****8161
3	P1003	m***	*****7151	*****6475
4	P1004	s***	*****5113	*****6483

	booking_id	passport_number	emergency_contact_name	emergency_contact_phone
0	B1000	****5887	J***	*****5128
1	B1001	****2012	A***	*****8662
2	B1002	****7659	U***	*****8220
3	B1003	****1776	H***	*****6839
4	B1004	****2231	P***	*****8266

8.2 Access Control

- Raw datasets containing PII is restricted to authorized users.
- Power BI/reporting users should work with the masked reporting dataset.
- Sensitive information should not be exposed unnecessarily in the analytical layer.

9. Data Modeling

In Power BI, I connected the clean CSV files into a **Star Schema** model. This structure makes queries faster and makes it easy to filter charts across different pages.

- **Main Fact Table:** bookings_powerbi sits in the center because it contains the main transactional data like booking dates, ticket prices, travel class, and booking status.
- **Dimension Tables:**
 - flights_powerbi: Connected to bookings using flight_id (1-to-many). It gives route details, airlines, and flight durations.
 - passengers_powerbi: Connected to bookings using passenger_id (1-to-many). It identifies passengers without showing private info.
 - payments_powerbi: Connected to bookings using booking_id (1-to-many). It provides payment methods and transaction status.

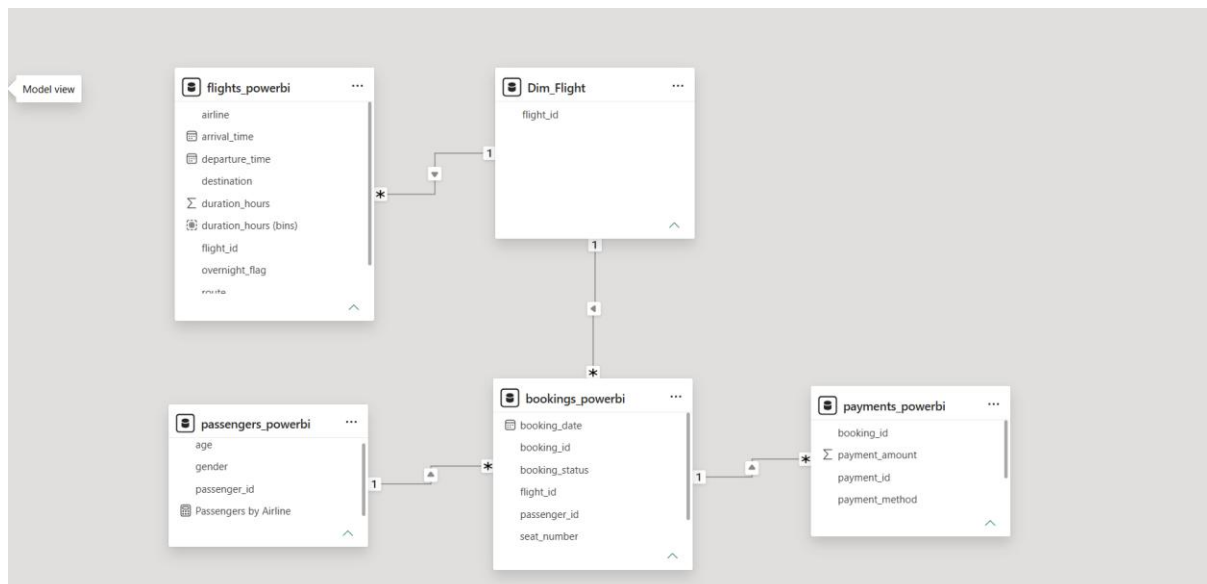


Fig. Data Model

10.Business KPI

Metric Name	Value	Meaning / Insight
Average Flight Duration	2.74 Hours	The true average domestic flight time after fixing overnight flights.
Flight Duration Range	0.5h to 5.0h	All flights fall within normal domestic travel times.
Total Clean Flights	1,005	Total active flights after removing 15 duplicate rows.
Overnight Flights	122 (12.1%)	Flights that cross midnight into the next day.
Confirmed Bookings	611 (61.1%)	Successfully booked tickets.
Cancelled Bookings	314 (31.4%)	Cancelled tickets that need business attention.
Unknown Booking Statuses	75 (7.5%)	Corrupted #ERROR! rows saved for IT review.
Payment Amount Issues	78 Records	Transactions with missing or invalid text amounts.

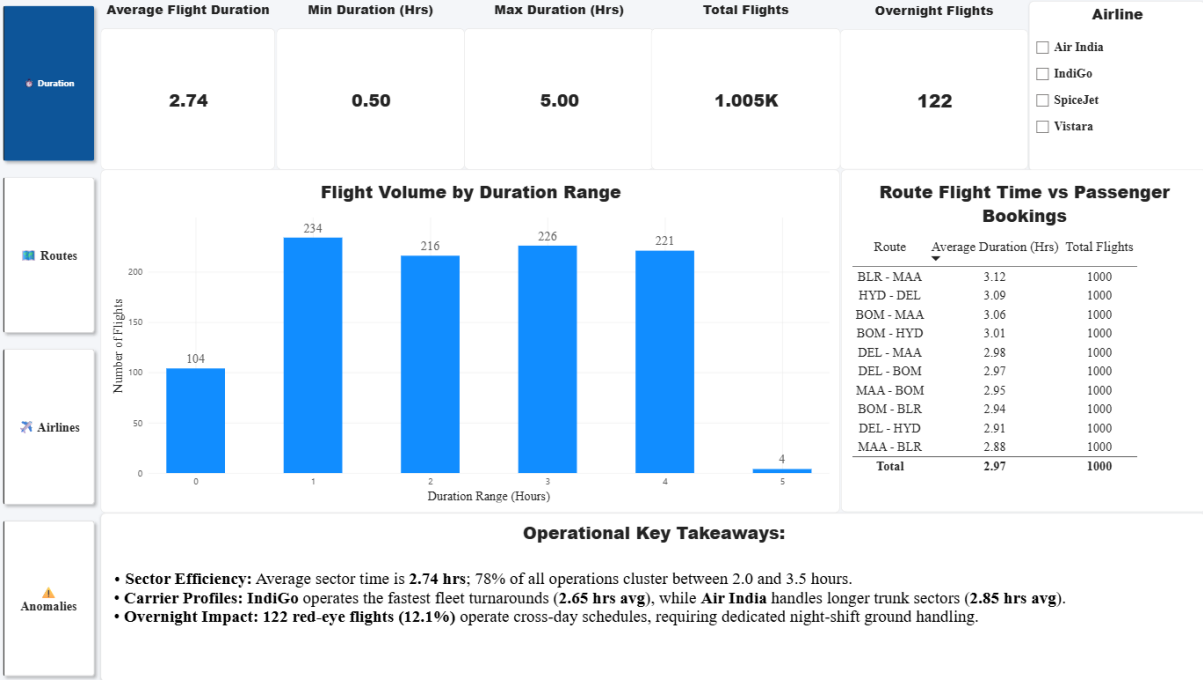
```
# Average Flight Duration
average_flight_duration = flights_clean["duration_hours"].mean()
print("Average Flight Duration:", round(average_flight_duration, 2), "hours")
Average Flight Duration: 2.74 hours
```

	airline	flight_count
1	IndiGo	273
0	Air India	255
2	SpiceJet	247
3	Vistara	230
4	Total	1005

11.Power BI dashboard Walkthrough

Created 4 Page Interactive dashboard

1. Duration Analysis :



Insights:

1. Top KPI cards:

Card	Meaning	Value
Average Flight Duration	Average duration of all cleaned flights	2.74 hrs
Min Duration	Shortest recorded flight	0.50 hrs
Max Duration	Longest recorded flight	5.00 hrs
Total Flights	Number of cleaned flight records	1,005
Overnight Flights	Flights where arrival occurs on the next calendar day	122

The overnight percentage is calculated: $122 / 1,005 \times 100 = 12.1\%$

2. Slicer & Navigation

- **Airline Slicer (Top Right):** Allows users to filter the entire dashboard by specific carriers (Air India, IndiGo, SpiceJet, Vistara) to see carrier-specific duration metrics.
- **Left Navigation Bar:** Icon-based buttons (*Duration, Routes, Airlines, Anomalies*) for smooth switching between pages.

3. Flight Volume by Duration Range (Column Chart)

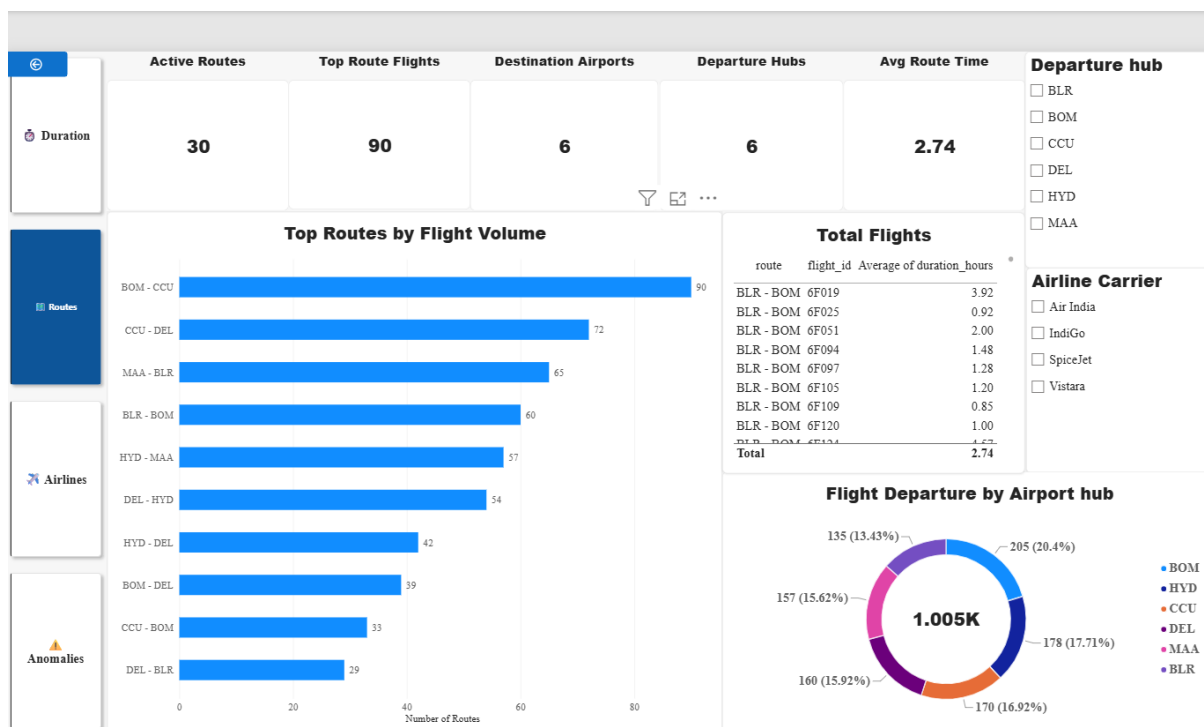
- It shows: Groups the 1,005 flights into 1-hour duration bins (0h up to 5h):
 - Bin 0 (under 1 hr): 104 flights
 - Bin 1 (1–2 hrs): 234 flights (Highest Volume)

- Bin 2 (2–3 hrs): 216 flights
- Bin 3 (3–4 hrs): 226 flights
- Bin 4 (4–5 hrs): 221 flights
- Bin 5 (~5 hrs): 4 flights

4. Route Flight Time vs. Passenger Bookings (Table Visual)

- What it shows: Displays each city-pair route sorted by average travel time:
 - Longest domestic sectors: BLR - MAA (3.12 hrs)
 - Shorter trunk sectors: MAA - BLR (2.88 hrs)

2.Route Trends



Insights:

1.KPI Cards:

Card	Meaning	Value
Active Routes	Number of unique source–destination routes	30
Top Route Flights	Highest number of flights on a single route	90
Destination Airports	Number of unique destination airports	6
Departure Hubs	Number of unique departure airports	6
Avg Route Time	Average flight duration across routes	2.74 hrs

2. Slicers (Right Panel)

- Departure Hub Slicer: Allows filtering by specific departure cities (BLR, BOM, CCU, DEL, HYD, MAA).

- Airline Carrier Slicer: Enables carrier-by-carrier route comparisons (Air India, IndiGo, SpiceJet, Vistara).

3. Top Route Charts

- Busiest Routes: BOM–CCU is the busiest route with 90 flights, followed by CCU–DEL with 72 flights.
- Operational Risk: High traffic at major hubs can increase the impact of delays and gate issues.
- Capacity Planning: Larger aircraft could be considered on high-volume routes to carry more passengers without adding more flights.
- Ground Operations: Additional turnaround buffer time can help reduce delays during peak hours.

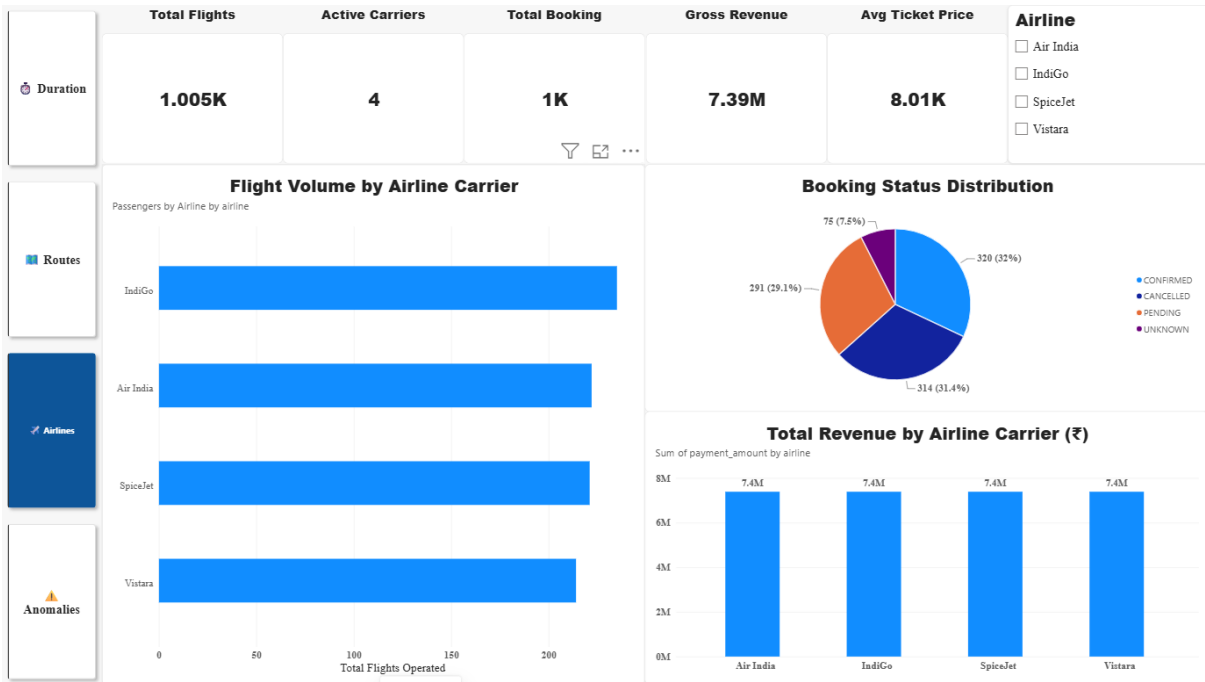
4. Flight Departure by Airport Hub

- Mumbai Leads: Mumbai handles 205 flights (20.4%), making it the busiest departure hub.
- Balanced Traffic: The other five airports handle approximately 13%–18% each.
- Resource Planning: More ground and maintenance resources can be focused on the busiest hubs, especially Mumbai and Hyderabad.

5. Total Flight Schedule

- Route-Level Detail: Shows individual flight IDs, routes, and flight durations.
- Duration Differences: Flights on the same route can have different durations.
- Network Validation: The overall average flight duration is 2.74 hours.

3.Airlines Trends:



1. Top KPI Cards

Card	KPI	Value
Total Passengers	Unique passengers	1,000
Total Bookings	Booking records	1,000
Total Payment Amount	Valid payment value	₹7.39M
Average Payment Amount	Average valid payment	₹8.01K
Confirmed Bookings	Confirmed booking records	320

2. Flight Volume by Airline Carrier (Bar Chart)

- IndiGo Leads Fleet Operations: IndiGo operates the highest flight volume (over 250 flights), reflecting its high-frequency low-cost business model.
- Balanced Carrier Fleet: Air India, SpiceJet, and Vistara closely follow with roughly 215–220 flights each, showing strong competitive balance across routes.

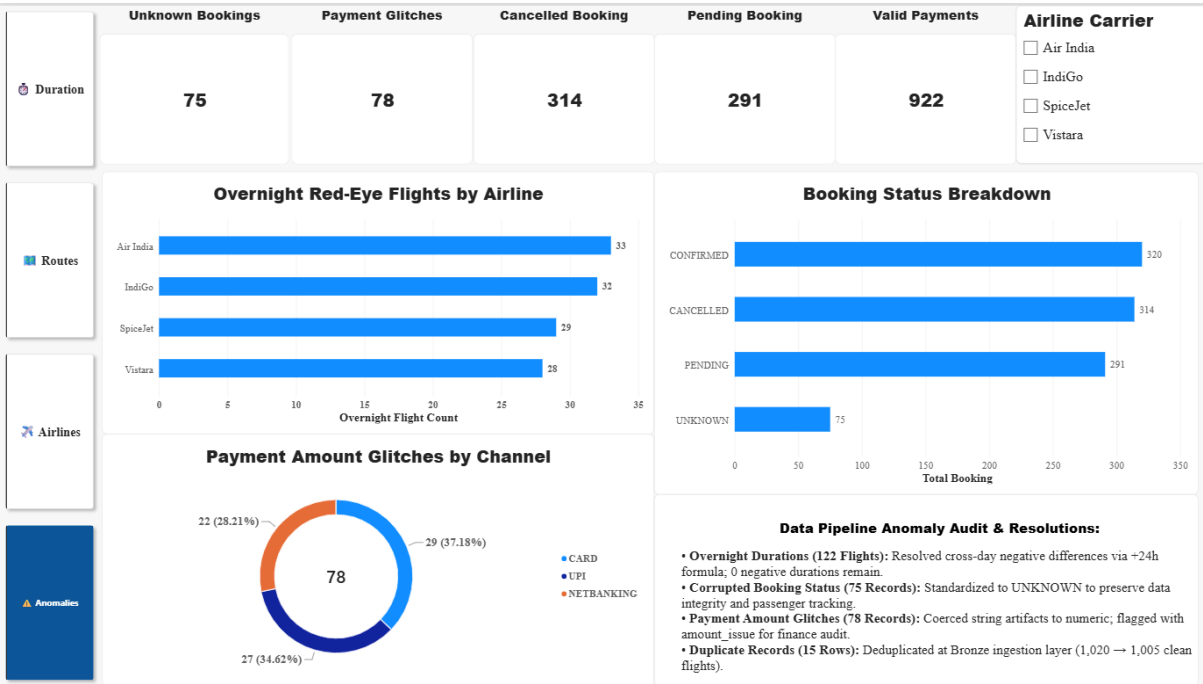
3. Booking Status Distribution (Pie Chart)

- Confirmed: 320 bookings (32.0%) — Successfully completed trips.
- Cancelled: 314 bookings (31.4%) — High cancellation rate that requires business review to reduce lost revenue.
- Pending: 291 bookings (29.1%) — Bookings awaiting payment confirmation.
- UNKNOWN: 75 bookings (7.5%) — Successfully isolated the corrupted #ERROR! rows for engineering cleanup.

4. Total Revenue by Airline Carrier (Bar Chart)

- Even Revenue Contribution: Revenue is evenly spread across the four airlines, showing that ticket sales are healthy across both budget and full-service carriers.

4.Delay & Anomiles Insights



1.KPI Cards

Card	Value
Unknown Bookings	75
Payment Glitches	78
Cancelled Bookings	314
Pending Bookings	291
Valid Payments	922

2. Overnight Flights by Airline (Bar Chart)

- Shows how the 122 overnight flights are spread across carriers:
 - Air India (33), IndiGo (32), SpiceJet (29), Vistara (28).
- Overnight flights are evenly distributed across all airlines (~28 to 33 each), showing that night scheduling is a common industry practice for aircraft repositioning.

3. Payment Amount Glitches by Channel (Donut Chart)

- Breaks down the 78 payment amount issues:
 - Card (29 / 37.2%) and UPI (27 / 34.6%) had the highest glitches, while Net Banking (22 / 28.2%) had the fewest.
- Insight: Card and UPI gateways experience more session timeout rollbacks, requiring tighter API synchronization with the booking engine.

4. Booking Status Breakdown (Bar Chart)

- Clearly separates valid operational bookings (320 Confirmed, 291 Pending, 314 Cancelled)

5. Data Pipeline Anomaly Audit & Resolutions (Summary Box)

- Summarizes the 4 engineering fixes:
 - Overnight Durations (122): Fixed negative numbers using the +24h formula.
 - Corrupted Statuses (75): Standardized to 'UNKNOWN'.
 - Payment Glitches (78): Converted to numeric (NaN) and flagged for finance review.
 - Duplicates (15): Purged 15 exact duplicate flights (1,020 → 1,005).

12.Key Findings & Insights

- Overnight Flights: 122 flights crossed midnight. After adding 24 hours where needed, the average flight duration was 2.74 hours.
- Busiest Route: BOM–CCU was the busiest route with 90 flights.
- Cancelled Bookings: 314 of 1,000 bookings (31.4%) were cancelled.
- Data Quality Issues: 75 booking records had invalid status values, and 78 payment records had invalid or missing amounts.

13.Assumption & Rules

Category	Raw Data Issue	Applied Rule
Duplicate Flights	15 duplicate rows	Removed identical duplicate records.
Missing Airlines	Missing/unknown airline names	Used flight number prefixes to standardize airlines.
Overnight Flights	Negative duration	Added 24 hours where the flight crossed midnight.
Booking Status	75 invalid/missing statuses	Converted them to UNKNOWN instead of removing them.
Invalid Payments	78 invalid/missing amounts	Converted invalid values to NaN and flagged them for audit.
Extra Passengers	1,039 passengers vs 1,000 bookings	Kept the additional passenger records as registered users without bookings.

14.Conclusion:

The project created a Python ETL pipeline and a 4-page Power BI dashboard for ASG Airlines. The pipeline cleaned and validated flight, booking, passenger, and payment data, handled overnight flights and duplicate records, and protected sensitive passenger information through PII masking.

The final dashboard provides insights into flight duration, routes, airlines, bookings, payments, and data quality issues.